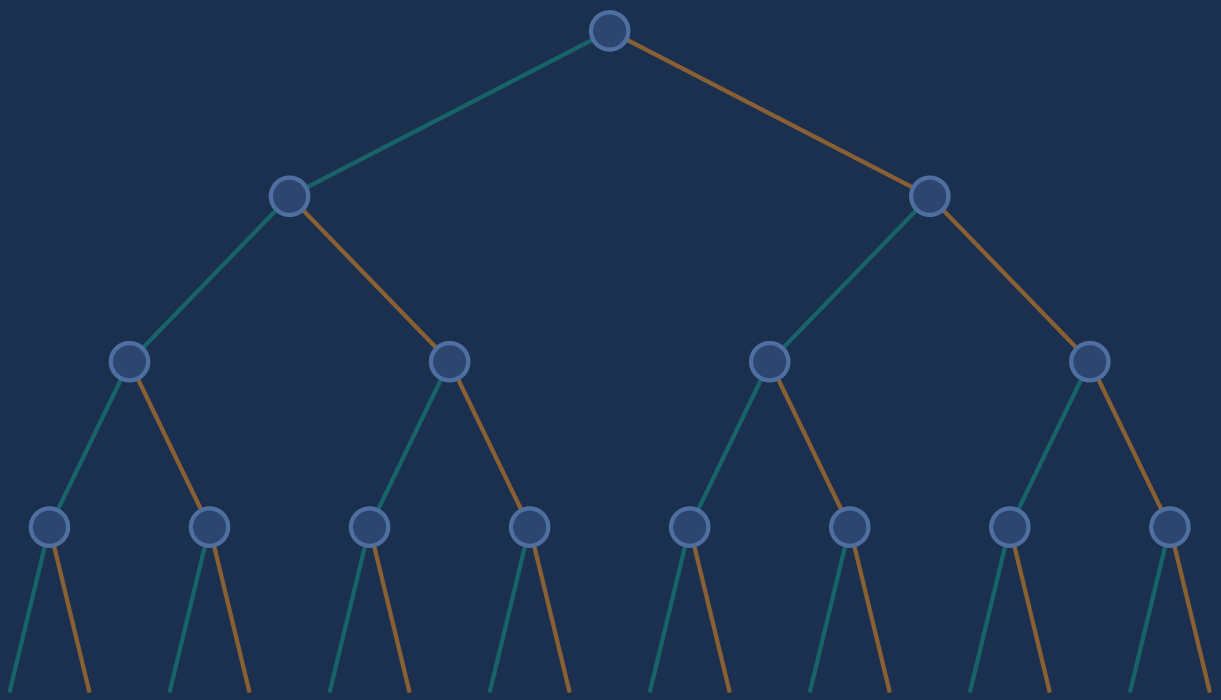
MULTIMEDIA

Shannon-Fano Algorithm

Section No. 5



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How Shannon-Fano Coding Works

Computers normally store **every character with 8 bits** (ASCII), no matter how often it appears. Shannon-Fano is a **lossless compression** method: characters that appear **often get short codes** and characters that appear **rarely get longer codes**, so the whole message needs fewer bits.

The 5 steps

1	**Count and sort** Count each symbol and sort the list from the most frequent to the least frequent.
2	**Split** Divide the list into two groups whose totals are **as equal as possible**.
3	**Label** Give the first (upper) group the bit **0** and the second (lower) group the bit **1**.
4	**Repeat** Repeat steps 2 and 3 inside every group that still has more than one symbol.
5	**Read the codes** The code of a symbol is the sequence of bits on the path from the root down to it.

Formulas we will use

Quantity	Formula	Meaning
Probability	p = count / total	How often the symbol appears
Ideal bits	$\log_2(1 / p)$	Best possible code length for the symbol
Subtotal bits	count x code length	Bits used by all copies of the symbol
Average bits	total bits / total symbols	Average code length per symbol
Entropy (H)	H = SUM of p x $\log_2(1 / p)$	Theoretical minimum average bits per symbol
Compression ratio	B0 / B1	B0 = symbols x 8 (before), B1 = total bits (after)

Good to know: the entropy is the smallest average code length that any code can reach. If the average of our code **equals** the entropy, the code is perfect. This happens when every probability is a power of 1/2 (1/2, 1/4, 1/8, 1/16 ...).

Warm-up: Coding the Word HELLO

GIVEN
Message: **HELLO** (5 characters)
ASCII uses 8 bits per character

FIND
Shannon-Fano code for each letter
Number of bits after compression
Compression ratio

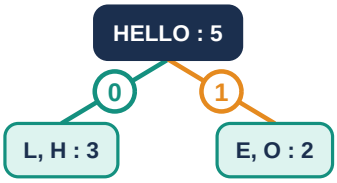
1 Count the letters, then split into groups with equal totals

L = **2**, H = **1**, E = **1**, O = **1** (total = **5** letters)

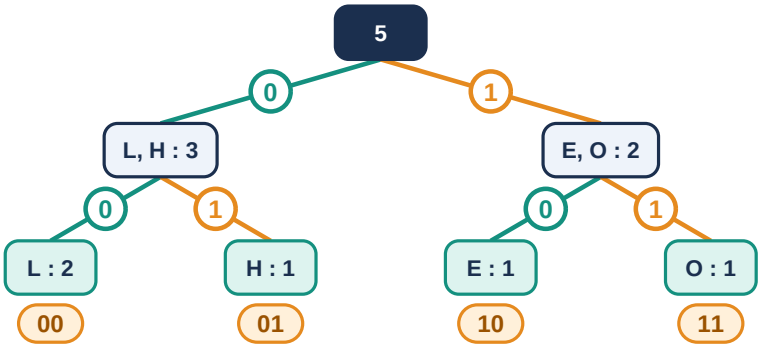
Split	Group	Total	Left group (bit 0)	Right group (bit 1)
1	L, H, E, O	5	L, H = 3	E, O = 2
2	L, H	3	L = 2	H = 1
3	E, O	2	E = 1	O = 1

2 Draw the tree

(a) after the first split



(b) the complete tree. Read each code from the top down.



3 Fill in the table

Symbol	Count	$\log_2(1/p)$	Code	Bits used (count x length)
L	2	1.32	00	2 x 2 = 4
H	1	2.32	01	1 x 2 = 2
E	1	2.32	10	1 x 2 = 2
O	1	2.32	11	1 x 2 = 2
TOTAL	**5**			**10 bits**

L: p = 2/5 = 0.4, so $\log_2(1/0.4) = 1.32$. H, E, O: p = 1/5, so $\log_2(5) = 2.32$.

4 Compression ratio

B0 = 5 x 8 = **40 bits** (before) | B1 = **10 bits** (after)
Compression ratio = B0 / B1 = 40 / 10 = **4**

Answer: L = 00, H = 01, E = 10, O = 11. Total = 10 bits. Compression ratio = 4.

Example 1: Eight Characters

QUESTION

Eight characters have the distribution A:(1), B:(1), C:(2), D:(4), E:(4), F:(4), G:(8), H:(8).
a) Draw a Shannon-Fano tree. b) Complete the table.
c) Calculate the average number of bits per character.
d) Calculate the entropy. e) Calculate the compression ratio.

1 Sort (most frequent first) and find the total

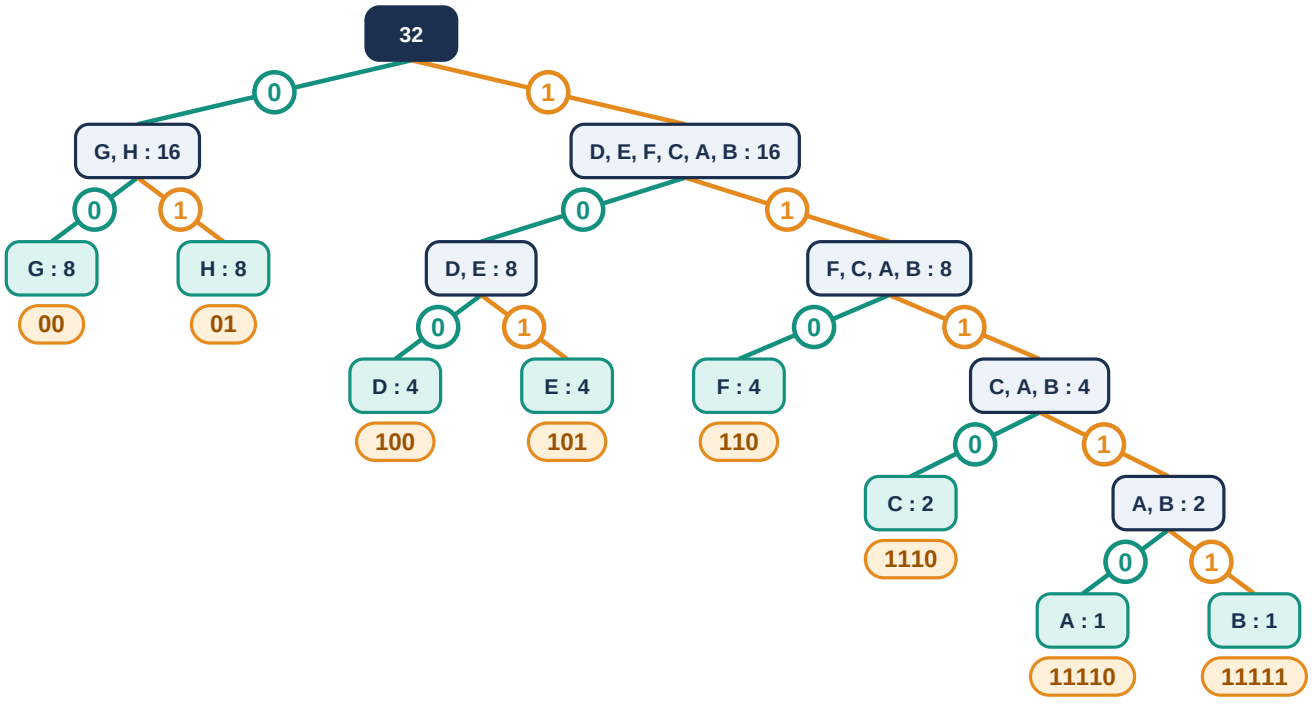
G = 8, **H** = 8, **D** = 4, **E** = 4, **F** = 4, **C** = 2, **A** = 1, **B** = 1

Total = 8 + 8 + 4 + 4 + 4 + 2 + 1 + 1 = **32 characters**

2 Split step by step (aim for equal totals)

Split	Group to divide	Total	Left (bit 0)	Right (bit 1)
1	G, H, D, E, F, C, A, B	32	G, H = 16	D, E, F, C, A, B = 16
2	G, H	16	G = 8	H = 8
3	D, E, F, C, A, B	16	D, E = 8	F, C, A, B = 8
4	D, E	8	D = 4	E = 4
5	F, C, A, B	8	F = 4	C, A, B = 4
6	C, A, B	4	C = 2	A, B = 2
7	A, B	2	A = 1	B = 1

3 Draw the Shannon-Fano tree



Teal = bit 0 (upper group) **Orange = bit 1** (lower group). The orange tag under each leaf is its code.

4 Complete the table

Symbol	Count	$\log_2(1/p)$	Code	Subtotal (# of bits)
G	8	2	00	8 x 2 = 16
H	8	2	01	8 x 2 = 16
D	4	3	100	4 x 3 = 12
E	4	3	101	4 x 3 = 12
F	4	3	110	4 x 3 = 12
C	2	4	1110	2 x 4 = 8
A	1	5	11110	1 x 5 = 5
B	1	5	11111	1 x 5 = 5
TOTAL	**32**			**86 bits**

Example: G has p = 8/32 = 1/4, so $\log_2(1/p) = \log_2(4) = 2$. A has p = 1/32, so $\log_2(32) = 5$.

5 Average number of bits per character

Average = total bits / total characters
= 86 / 32 = **2.6875 bits per character**

6 Entropy

Symbol	p	$\log_2(1/p)$	$p \times \log_2(1/p)$
G, H (each)	8/32 = 1/4	2	0.25 x 2 = 0.5
D, E, F (each)	4/32 = 1/8	3	0.125 x 3 = 0.375
C	2/32 = 1/16	4	0.0625 x 4 = 0.25
A, B (each)	1/32	5	0.03125 x 5 = 0.15625

H = (2 x 0.5) + (3 x 0.375) + 0.25 + (2 x 0.15625)
H = 1 + 1.125 + 0.25 + 0.3125 = **2.6875 bits**

The average (2.6875) **equals** the entropy (2.6875), so this code is perfect. All probabilities are powers of 1/2.

7 Compression ratio

B0 = 32 x 8 = **256 bits** (before compression)
B1 = **86 bits** (after compression)
Compression ratio = B0 / B1 = 256 / 86 = **2.98**

Answer: Average = 2.6875 bits/char, Entropy = 2.6875 bits, Compression ratio = 2.98.

Example 2: Gray-Level Image

QUESTION

What is the entropy of the image below? (the numbers 0, 20, 50, 99 are gray-level intensities)
Show step by step how to build the Shannon-Fano tree and give the code for each intensity.
What is the average number of bits per pixel? How does it compare to the entropy?
Calculate the compression ratio.

The image (8 x 8 = 64 pixels)

99	99	99	99	99	99	99	99
20	20	20	20	20	20	20	20
0	0	0	0	0	0	0	0
0	0	50	50	50	50	0	0
0	0	50	50	50	50	0	0
0	0	50	50	50	50	0	0
0	0	50	50	50	50	0	0
0	0	0	0	0	0	0	0

Colour key

0	darkest
20	dark
50	medium
99	brightest

1 Count how many pixels have each intensity

Rows	0	20	50	99
Row 1				8
Row 2		8		
Row 3	8			
Rows 4 to 7 (4 rows)	4 x 4 = 16		4 x 4 = 16	
Row 8	8			
TOTAL	**32**	**8**	**16**	**8**

Check: 32 + 8 + 16 + 8 = **64 pixels**

2 Probabilities and entropy

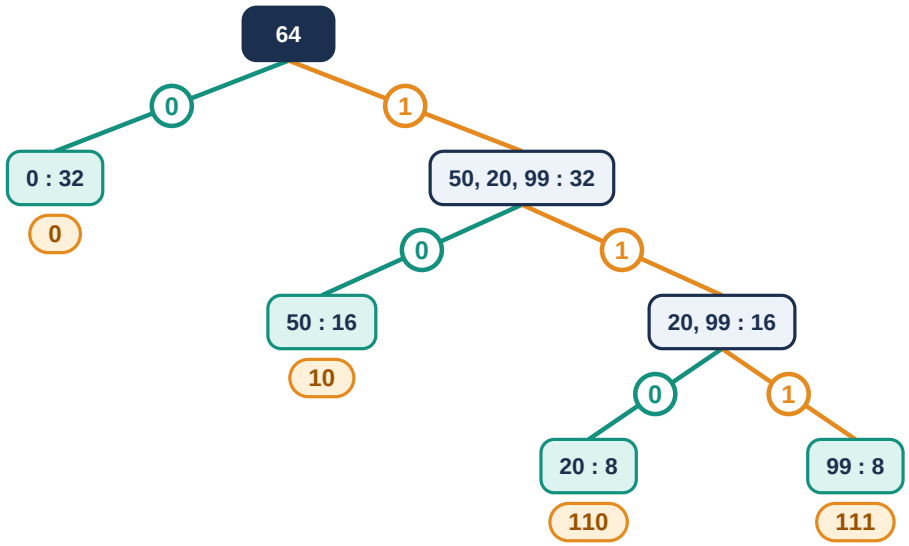
Intensity	Count	p = count/64	$\log_2(1/p)$	p x $\log_2(1/p)$
0	32	1/2	1	0.5 x 1 = 0.5
50	16	1/4	2	0.25 x 2 = 0.5
20	8	1/8	3	0.125 x 3 = 0.375
99	8	1/8	3	0.125 x 3 = 0.375
TOTAL	**64**	**1**		**H = 1.75 bits**

H = 0.5 + 0.5 + 0.375 + 0.375 = **1.75 bits per pixel**

3 Build the Shannon-Fano tree step by step

Sorted list: **0 (32), 50 (16), 20 (8), 99 (8)**

Split	Group to divide	Total	Left (bit 0)	Right (bit 1)
1	0, 50, 20, 99	64	0 = 32	50, 20, 99 = 32
2	50, 20, 99	32	50 = 16	20, 99 = 16
3	20, 99	16	20 = 8	99 = 8



Teal = bit 0 **Orange = bit 1**

4 The resulting code for each intensity

Intensity	Count	Code	Code length	Bits used
0	32	0	1	32 x 1 = 32
50	16	10	2	16 x 2 = 32
20	8	110	3	8 x 3 = 24
99	8	111	3	8 x 3 = 24
TOTAL	**64**			**112 bits**

5 Average bits per pixel, compared with the entropy

Average = total bits / pixels = 112 / 64 = **1.75 bits per pixel**

Entropy = **1.75 bits per pixel**

The average **equals** the entropy, so Shannon-Fano reached the theoretical limit here (all probabilities are powers of 1/2). For comparison, a fixed-length code would need 2 bits per pixel (4 values), i.e. 64 x 2 = 128 bits.

6 Compression ratio

B0 = 64 x 8 = **512 bits** (before compression)

B1 = **112 bits** (after compression)

Compression ratio = B0 / B1 = 512 / 112 = **4.57**

Answer: Entropy = 1.75 bits. Codes: 0 = "0", 50 = "10", 20 = "110", 99 = "111". Average = 1.75 bits/pixel (equal to the entropy). Compression ratio = 4.57.

Summary of Results

Problem	Total bits (B1)	Average bits	Entropy	Compression ratio
HELLO	10	2	1.92	4
Example 1	86	2.6875	2.6875	2.98
Example 2	112	1.75	1.75	4.57

Remember: (1) sort by count, (2) split into two nearly equal groups, (3) 0 for the first group and 1 for the second, (4) repeat until every symbol is alone. Average bits can never be smaller than the entropy.